

Landslides

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Stage-based deformation preceding rainfall-induced landslide failure in Songxi County, China

Abstract Rainfall-induced landslides are common hazards in the hilly regions of southern China, yet detailed observations of deformation evolution during rainstorm events and its relationship with rainfall-derived hydrological forcing remain limited. This study investigates two rainfall-induced landslides that occurred in June 2024 in Songxi County, China, using remote sensing, field surveys, in situ monitoring, geotechnical investigations, and laboratory testing. Continuous monitoring at Landslide No. 1 enabled comparison between a significant deformation episode in 2022 and the subsequent failure in 2024. The results reveal a stage-based deformation process, including an initial response stage, a sustained deformation phase, and a final acceleration stage. By combining effective rainfall, used as a rainfall-derived proxy for antecedent wetness, with displacement monitoring, the transition from sustained deformation to rapid acceleration could be identified before failure. Field observations suggest that anthropogenic disturbances, including slope cutting, drainage degradation, and vegetation change, may have increased local slope susceptibility by modifying runoff concentration, drainage efficiency, and infiltration pathways. These findings indicate that failure was associated not only with rainfall magnitude but also with antecedent rainfall accumulation, deformation state, and local slope conditions, highlighting the value of integrating rainfall-derived indicators with displacement monitoring for early warning.

Keywords Colluvial landslides · Extreme rainfall events · Field investigation · Human activities · Early warning

Introduction

Extreme rainfall events have intensified in both frequency and magnitude under ongoing climate change, significantly increasing the occurrence of rainfall-induced landslides and associated risks to infrastructure and communities (Marchi et al. 2010; Gobiet et al. 2014; Cappelli et al. 2021; Dunn et al. 2022). Rainfall is widely recognized as the dominant trigger of non-seismic landslides, accounting for approximately 85% of documented cases worldwide (Haque et al. 2019). Future projections further indicate a continued increase in landslide activity under intensified precipitation regimes (Gariano and Guzzetti 2016; NASA 2020).

The response of slopes to rainfall is governed not only by total precipitation but also by the combined effects of rainfall intensity and duration, which control infiltration, pore-water pressure development, and shear strength reduction (Ran et al. 2015, 2018; Tang et al. 2018; Wadhawan et al. 2020; Sato and Shuin 2024). However, slope response is inherently non-linear and strongly dependent on site-specific conditions (Matsuura et al. 2008; Bogaard and Greco

2015; Sidle and Bogaard 2016). Local factors such as topography, lithology, and subsurface structure modulate hydrological processes and can produce complex, time-lagged responses, meaning that similar rainfall events may lead to markedly different slope behaviors (Montgomery and Dietrich 1994; Van Asch et al. 1999; Rosso et al. 2006). Empirical rainfall thresholds have been widely used for landslide early warning by relating antecedent and event rainfall to failure occurrence (Baum and Godt 2009; Sengupta et al. 2009; Kirschbaum et al. 2011). However, their predictive capability is often limited, as they do not account for the evolving internal state of slopes (Thomas et al. 2018; Mirus et al. 2025). Progressive deformation, hydrological conditioning, and structural adjustments can significantly alter slope sensitivity over time, leading to markedly different responses under similar rainfall forcing (Bontemps et al. 2020; Song et al. 2022). Moreover, local site conditions can further modulate hydrological pathways and slope response, contributing to the observed variability in landslide occurrence (Bogaard and Greco 2015; Sidle and Bogaard 2016). As a result, slopes that have experienced prior deformation or structural disturbance may exhibit enhanced sensitivity, even when subjected to comparable rainfall inputs (Leshchinsky et al. 2021; Handwerger et al. 2022). This highlights the importance of considering the temporal evolution of slope conditions rather than relying solely on rainfall-based metrics (Bogaard and Greco 2018; Jones et al. 2021; Mirus et al. 2025).

Against this background, this study documents two rainfall-induced landslides, Landslides No. 1 and No. 2, which occurred on 15–16 June 2024 in Wushantou Village, Songxi County, Fujian Province, China. Only Landslide No. 1 exhibited significant deformation during a rainfall event in 2022. Both landslides were triggered during the same rainfall episode in June 2024. This contrast provides a useful case for examining how slope response evolves over time and exploring the factors associated with the transition from deformation to failure. The objectives of this study are to (i) characterize the geomorphological setting, rainfall conditions, and landslide features; (ii) analyze the evolution of slope response from deformation to failure, with particular emphasis on the contrasting behaviors of Landslide No. 1 observed in 2022 and 2024; and (iii) discuss implications for landslide early warning based on the combined integration of rainfall information and deformation monitoring.

Study area

Overview of the landslide events

The studied landslides occurred in Wushantou Village, Songxi County, Nanping City, Fujian Province, China, an area

characterized by steep hilly terrain and a subtropical monsoon climate. The landscape is predominantly mountainous, with farmland and water bodies accounting for the remaining approximately 20% of the area. Dense vegetation cover and abundant rainfall, especially during the plum rain season in June, provide favorable conditions for rainfall-induced slope instability. Figure 1 shows the regional setting, geological background, and the locations of Landslides No. 1 and No. 2.

The landslide sites have been strongly modified by human activities. Residential houses were constructed on stepped slopes, while road excavation created multiple cut slopes and benches. In addition, inadequate maintenance of intercepting ditches and roadside drainage facilities promoted local water accumulation

before failure. Several drainage structures were either blocked or damaged, allowing runoff to infiltrate into the slope through surface cracks and disturbed zones, thereby increasing the likelihood of localized saturation.

Between 9 and 18 June 2024, Songxi County experienced a prolonged heavy rainfall event with a cumulative precipitation of approximately 550 mm. The maximum daily rainfall reached 193.5 mm on 15 June. On the same day, Landslide No. 1 showed rapid deformation, and a yellow warning alert was issued. Landslide No. 2 occurred the following morning. Together, the two landslides destroyed 18 houses and partially damaged another 7, resulting in direct economic losses of approximately 25 million CNY. Owing to timely emergency warning and evacuation, no casualties were reported.

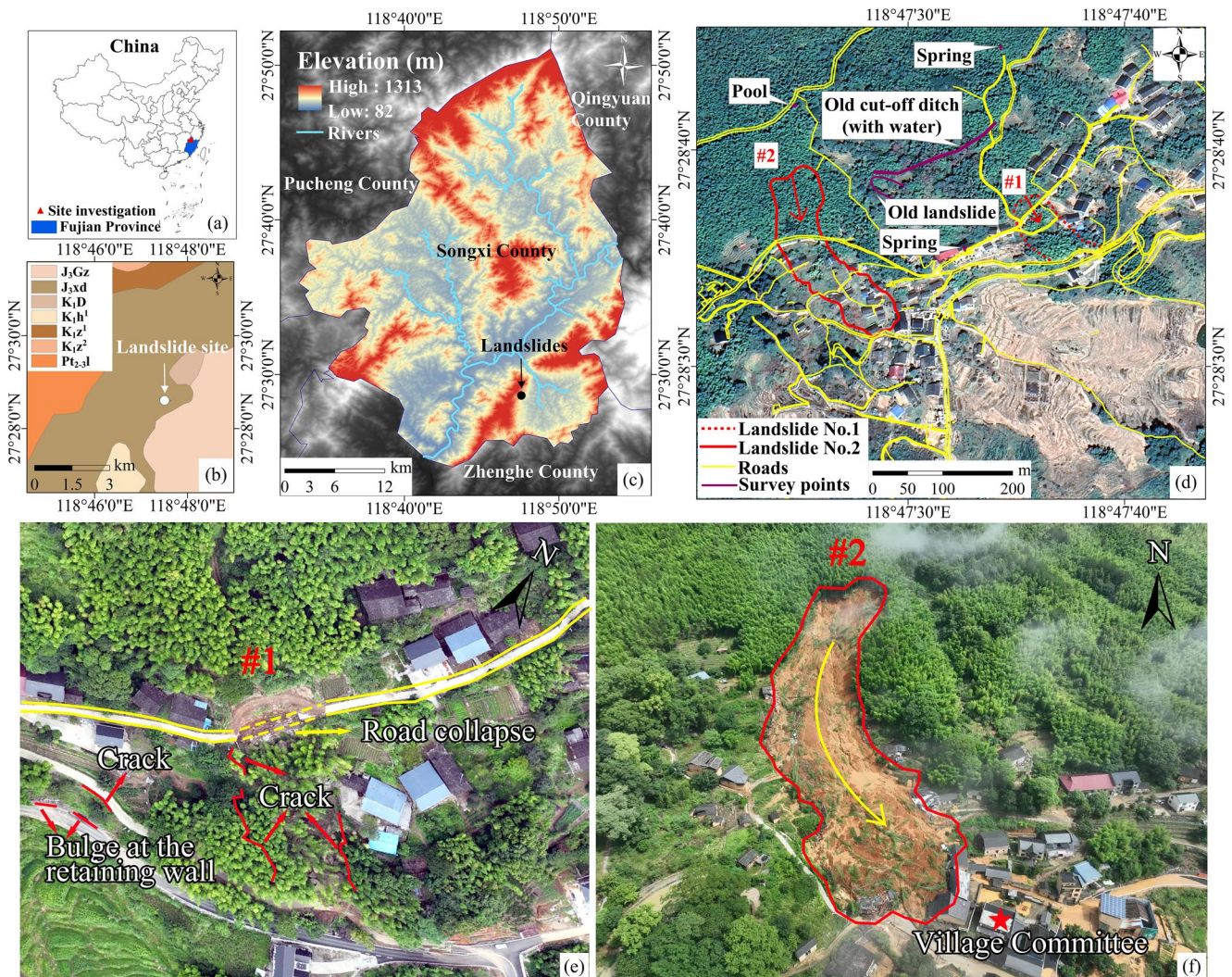


Fig. 1 Location of the Songxi landslides. **a** Geographic location of Songxi County within Fujian Province, China; **b** Geological map of the study area (scale 1:50,000); **c** Topography of Songxi County and locations of Landslides No. 1 and No. 2; **d** Aerial view of the landslide sites and surrounding human disturbances; **e** Post-failure UAV image of Landslide No. 1; and **f** Post-failure UAV image of Landslide No. 2

Topographic, geological, and rainfall conditions

The landslides are located on the southern piedmont slope of Mount Zhanlu, where the elevation reaches approximately 1230 m. The hillslope exhibits an overall concave morphology, with steep upper slopes of 40–60°, intermediate slopes of 30–40°, and relatively gentle lower slopes of 15–25°. Vegetation is well-developed in the middle and lower parts of the slope, where artificial bamboo plantations (*Phyllostachys edulis*) dominate, whereas the upper slope is characterized by exposed bedrock and patches of natural forest.

The geological setting is dominated by Late Yanshanian granodiorite and Jurassic tuff of the Xiadu Formation. The slope materials mainly comprise residual-deluvial clay, gravelly soil, and highly weathered bedrock. No active faults have been identified in the immediate vicinity of the study area. The area has a typical subtropical humid monsoon climate, with most rainfall concentrated between March and August. Long-term rainfall records from 1998 to 2023, obtained from the Fujian Meteorological Bureau, were used to characterize the regional rainfall regime. As shown in Fig. 2, the highest mean monthly precipitation occurs in May and June, corresponding to the plum rain season. In June 2024, the monthly rainfall reached 716.4 mm, which was nearly twice the long-term average for the same month.

Methodology

Data acquisition and landslide characterization

A combination of remote sensing, field surveys, in situ monitoring, geotechnical investigations, and laboratory testing was employed to comprehensively characterize the two landslides and to analyze the rainfall-deformation response. Remote-sensing data were used to document pre- and post-failure topographic and geomorphological conditions. UAV surveys were conducted using a DJI Mavic 3 M equipped with Real-Time Kinematic (RTK) positioning, generating post-failure orthophotos with a spatial resolution of 0.15 m and digital elevation models (DEMs) from a flight altitude of 400 m. Pre-failure imagery from November 2020 was obtained from Google

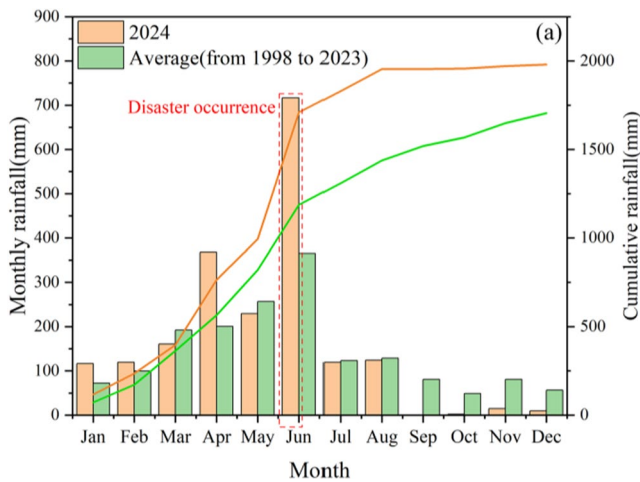


Fig. 2 The historical average monthly rainfall and the average monthly rainfall in 2024

Earth, while topographic data were derived from the JAXA ALOS dataset (12.5 m resolution).

Field surveys documented geomorphological changes, surface cracks, failure boundaries, and landslide characteristics. Following the initial signs of slope instability in March 2022, a monitoring system was installed at the Landslide No. 1 site (Fig. 3). The system consisted of two GNSS stations, four tiltmeters, one creepmeter, and a rain gauge. These instruments provided continuous records of rainfall and slope deformation prior to the June 2024 failure event. Specifically, the rain gauge was a WRS-01 piezoelectric rain gauge, with a measurement range of 0–8 mm/min, a resolution of 0.2 mm, a manufacturer-specified measurement accuracy of $\pm 4\%$, and an effective piezoelectric panel size of 200 mm. It was installed at 118.793460°E and 27.476155°N, on a level and stable support in an open area with no immediate overhead obstruction. Routine inspections were conducted to maintain normal operation of the monitoring sensor. Rainfall records were exported from the monitoring platform at an hourly resolution and then aggregated into daily and cumulative rainfall values.

Furthermore, geotechnical investigations, including borehole drilling, electrical resistivity surveys, and surface wave testing, were jointly used to constrain the stratigraphic configuration of the landslide area. These investigations provided information on the spatial distribution and thickness variation of the colluvial deposits, residual soils, fully weathered materials, and weathered bedrock. Laboratory testing was used to obtain the saturated hydraulic conductivity of the principal surficial layers. The integrated interpretation of these datasets provided the geological basis for the cross-sections shown in Fig. 4 and supported the interpretation of landslide geometry used in the rainfall-deformation analysis.

Rainfall indices and episode separation

Following the conceptual distinction between rainfall events and rainfall episodes proposed by Lu et al. (2024), the rainfall time series was organized into rainfall events and rainfall episodes. To focus on rainfall periods with clearer hydrological relevance, a rainfall event was defined as a sequence of consecutive rainfall days with daily rainfall exceeding 5.0 mm/day. A rainfall episode was then defined as a sequence of rainfall events separated by inter-event intervals shorter than T_c , thereby representing a continuous period of antecedent wetting and cumulative rainfall influence on the slope.

The parameter T_c controls the separation or merging of rainfall events into rainfall episodes. Instead of prescribing T_c only on the basis of a fixed empirical interval, a first-order hydraulic-timescale estimate was used as a reference for rainfall-episode separation (Lu et al. 2024). The characteristic timescale was estimated as:

$$T_c = \sum_{j=1}^m \frac{H_j}{K_{s,j}} \quad (1)$$

where H_j is the representative infiltration depth of the j th layer, $K_{s,j}$ is the corresponding saturated soil hydraulic conductivity, and m represents the number of the soil layers considered. This estimate assumes a simplified vertical flow path through the principal surficial layers and approximates the terminal seepage velocity using the saturated hydraulic conductivity under a

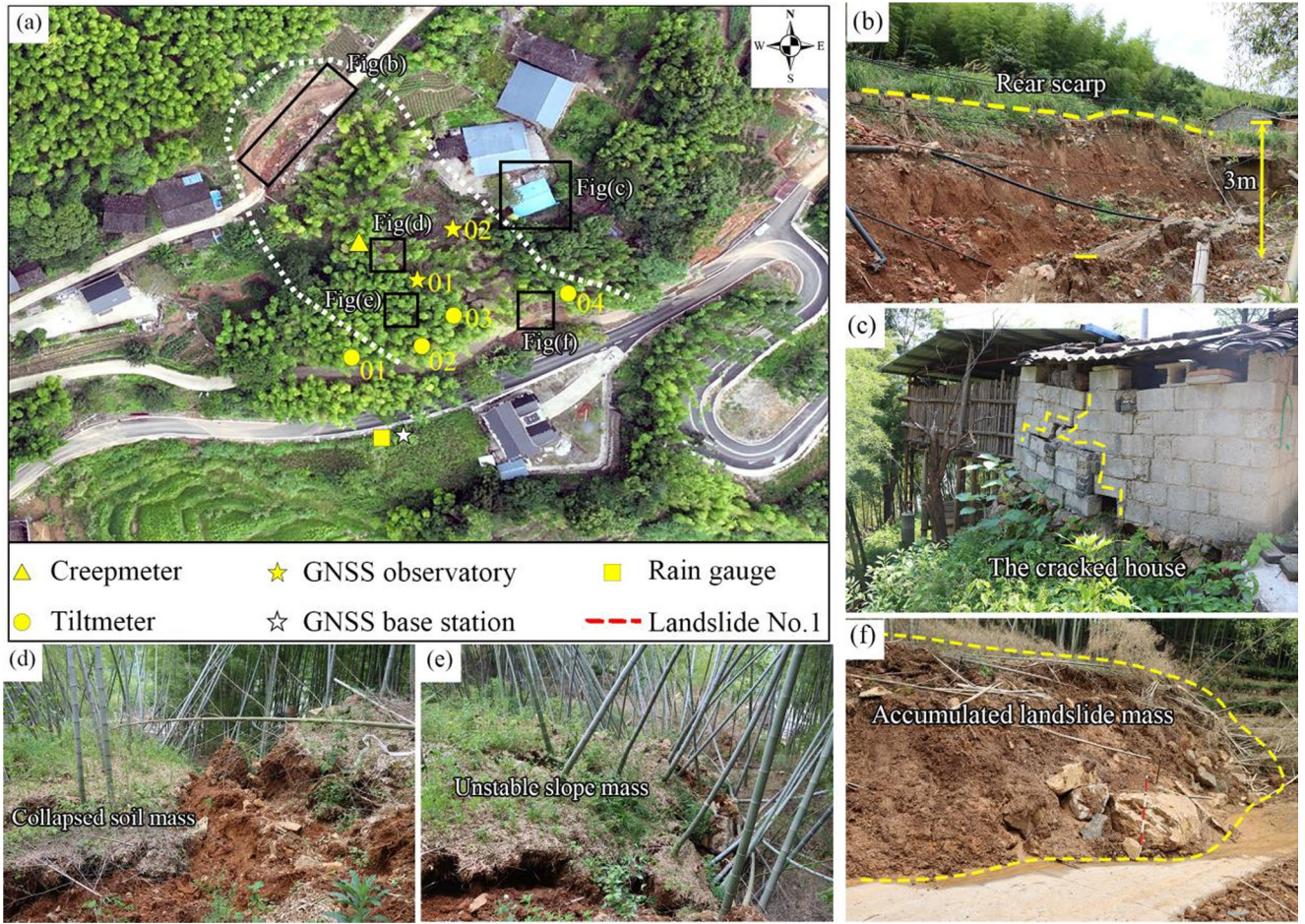


Fig. 3 The location of the in situ monitoring system and field photos at the site of Landslide No. 1 (photos taken in July 2024). **a** Locations of monitoring sensors, including two GNSS stations, one creepmeter, four tiltmeters, one rain gauge and early warning notification equipment; **b** The entire rear scarp subsided by 3 m; **c** damage to the nearby residential houses; **d** The collapsed soil mass due to shear failure; **e** Well-developed soil cracks and *Phyllostachys edulis* in a toppled state; and **f** Accumulated landslide mass blocking the road

simplified infiltration framework (Green and Ampt 1911; Lu and Godt 2013; Lu and Likos 2004).

For Landslide No. 1, the stabilized borehole water level obtained during the engineering-geological site investigation conducted in July 2024 was used as the reference depth for the T_c calculation (Iverson 2000; Collins and Znidarcic 2004). The observation point was located approximately 10 m from GNSS02, and the static water level was approximately 7.8 m below the ground surface. This value represents a single site-investigation observation rather than a continuously monitored groundwater condition during the 2022 and 2024 rainfall events. Hence, the simplified conceptual flow path included the upper colluvial gravelly soil layer (4.8 m thick, $K_s = 4.33 \times 10^{-3}$ cm/s) and the underlying residual sandy clay layer (3.0 m thick, $K_s = 1.05 \times 10^{-3}$ cm/s). The estimated layer-specific hydraulic timescale contributions were approximately 1.3 days and 3.3 days, respectively, yielding a total characteristic hydraulic timescale of approximately 5 days. A 5-day interval was therefore adopted as the rainfall-episode separation window, and rainfall events separated by less than 5 days were grouped into a single rainfall episode.

Both the current rainfall input and antecedent rainfall accumulation are widely recognized as important hydrological controls on rainfall-induced landslides (Rahardjo et al. 2001; Guzzetti et al. 2007; Abancó et al. 2024; Liang et al. 2025). Therefore, based on the identified rainfall episodes, an effective rainfall index (R_e) was estimated to represent the cumulative rainfall forcing and hydrological preconditioning (Crozier 1999; Glade 2000), which was defined as:

$$R_e = R_0 + \sum_{i=1}^n k^i R_i \quad (2)$$

where R_0 is the rainfall on the current day, R_i is the rainfall on the i^{th} day prior to the current day, k is a decay coefficient describing the rate of soil water dissipation, and n represents the number of daily time steps from the beginning of the corresponding rainfall episode to the calculation day (Glade 2000). Considering the humid climatic conditions of Southeast China, a decay coefficient of 0.8 was adopted (Di et al. 2019).

In this study, it should be noted that the transient unsaturated flow process, hydraulic gradients, preferential flow paths, and

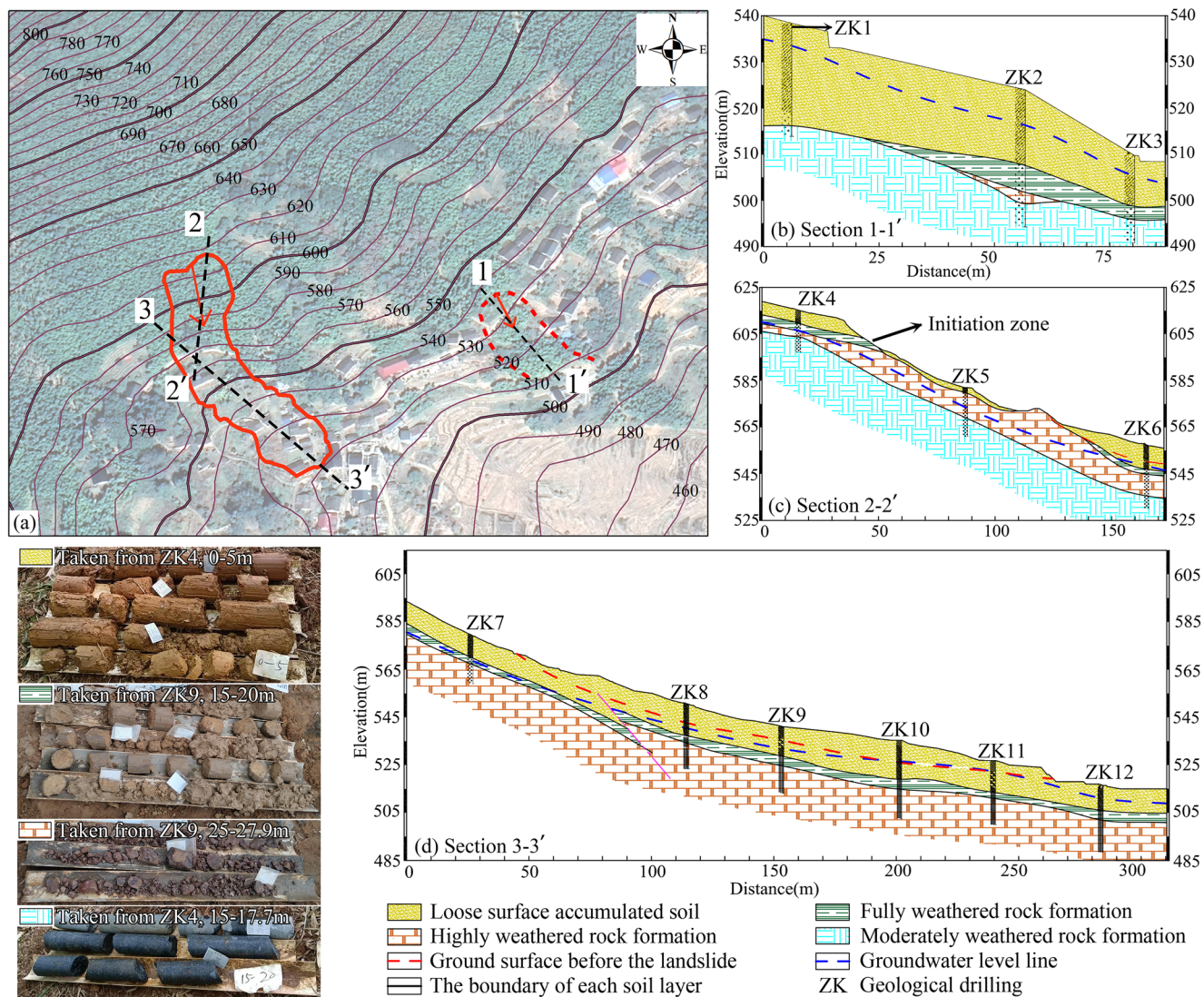


Fig. 4 Contour map of the studied area (a) and detailed geological profiles of sections 1-1' (b), 2-2' (c) and 3-3' (d)

transient infiltration behavior were not considered in the estimation of T_c . Moreover, the absence of continuous groundwater-level or pore-water-pressure monitoring may introduce uncertainty into the estimation of T_c . Hence, the estimated T_c represents a first-order, physically informed reference interval for rainfall-event separation, which should not be interpreted as a measured infiltration travel time or a direct estimate of groundwater or pore-pressure response. A sensitivity analysis was conducted using T_c values ranging from 1 to 7 days. Values of 1–3 days resulted in a more fragmented division of the rainfall events, whereas the classification of the principal rainfall sequences was broadly consistent for T_c values of 4–7 days. The estimated T_c of approximately 5 days falls within this stable range and was therefore adopted for rainfall-episode delineation. The principal temporal relationships between the identified rainfall episodes and displacement evolution were not materially affected within the tested range of 4–7 days. Moreover, R_c was used as an indirect proxy for antecedent hydrological conditions. It does not

directly measure subsurface hydrological processes, but provides a practical means of characterizing cumulative rainfall forcing and hydrological preconditioning when direct hydrological observations, e.g., pore-water pressure, soil-moisture, and matric suction, are unavailable.

Results

Landslide No. 1

Landslide No. 1 had been identified as a potential hazard site in 2022 and showed renewed deformation following early June rainfall in 2024, when new surface cracks and ground subsidence were observed. The landslide is located on a concave hillslope at the foot of Mount Zhanlu, with a contributing catchment area of approximately 0.07 km². The slope changes from a gentle upper section to a steeper lower segment (15–25°), and the failure occurred within a bamboo-forested area above a village road. The landslide exhibited

a tongue-shaped morphology, with an axial length of 88 m, a width of 30–45 m, and an estimated thickness of 3–6 m, corresponding to a volume of approximately $1.5 \times 10^4 \text{ m}^3$. The rear scarp showed vertical displacement of approximately 3 m, causing damage to the village road (Fig. 3b) and nearby buildings (Fig. 3c). Field observations documented extensive tensile cracking, toppled vegetation, and accumulation of displaced material at the slope toe (Fig. 3d–f).

Field investigations and borehole data revealed a four-layer stratigraphic structure with distinct physical and hydraulic characteristics (Fig. 4). The surficial loose layer consists of a heterogeneous mixture of fill, colluvial, and residual soils with weak cementation and heterogeneous thickness, influenced by both depositional processes and human activity. This layer is typically in a loose to moderately compacted state with a porosity of 0.42–0.49 and a saturated hydraulic conductivity of $0.96\text{--}4.33 \times 10^{-5} \text{ m/s}$. The fully

weathered layer is brownish-yellow and is composed of granodiorite (1.00–5.80 m thick) and tuff (2.20–5.00 m thick). Due to intense weathering, the original rock structure has been destroyed, and the primary minerals have largely transformed into secondary clay minerals. The material has a sand-like texture and is susceptible to softening upon wetting. The highly weathered layer is characterized by a largely destroyed rock structure, with locally preserved sand-like textures, dense fractures, and blocky fragments. The moderately weathered bedrock is light gray to black, with a banded to massive structure, and is mainly composed of relatively intact diorite with well-developed joints.

The temporal evolution of rainfall and surface displacement at Landslide No. 1 is shown in Fig. 5a. GNSS02, located within the main monitored deformation area, recorded significant deformation that was temporally associated with major rainfall events

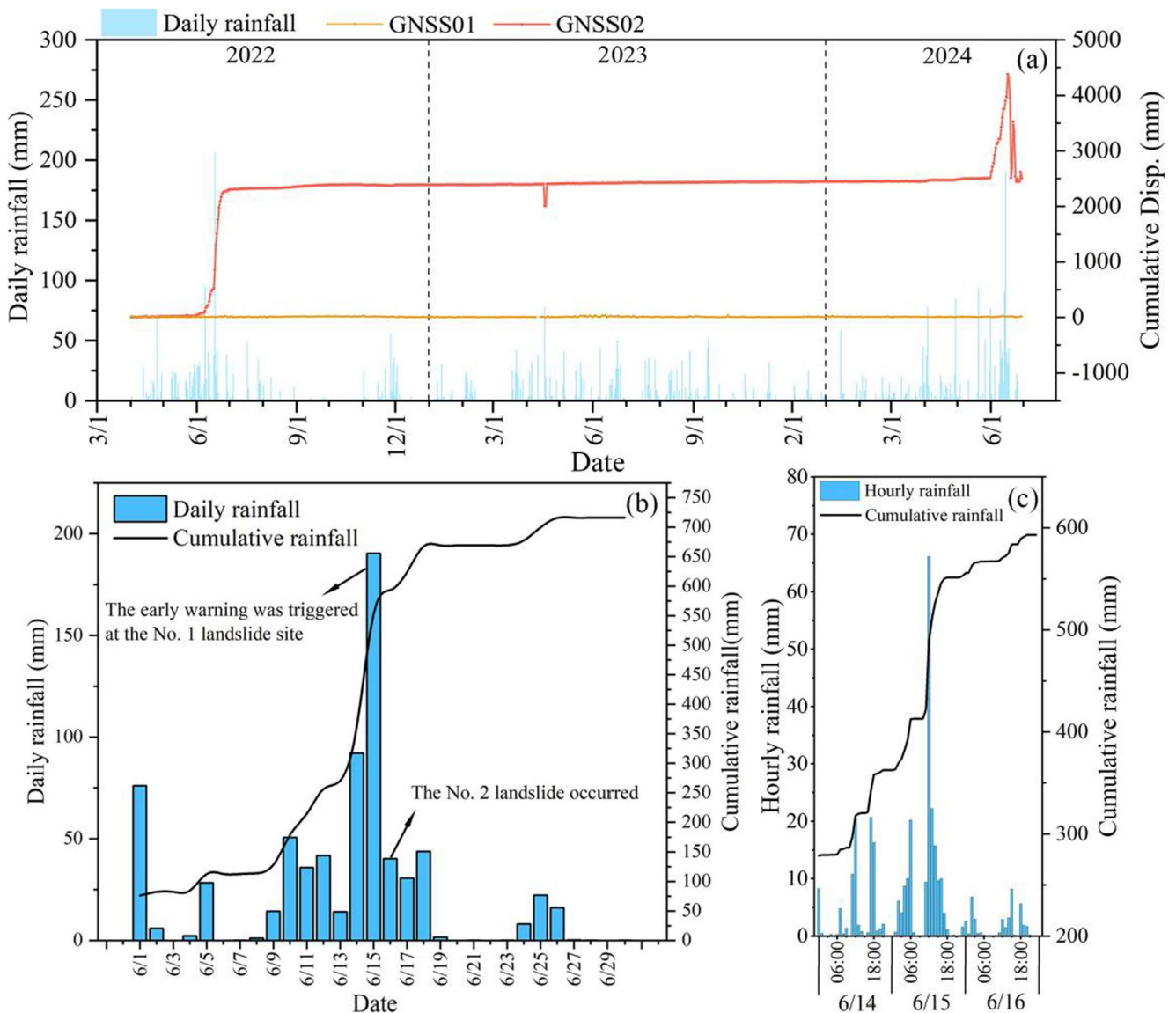


Fig. 5 Monitoring data at Landslide No. 1: **a** Time series of rainfall and ground displacement; **b** Daily rainfall during June 2024; and **c** Hourly rainfall before and after the landslide events in 2024

in both 2022 and 2024. On June 9, 2022, daily rainfall reached 94.4 mm, after which the displacement rate increased markedly. A subsequent extreme rainfall event on June 18, 2022 (206.4 mm/day) further accelerated the displacement rate, resulting in a cumulative displacement of 2.33 m. A similar response was observed in 2024, when intense rainfall episodes led to a cumulative displacement of 1.86 m between June 1 and June 17. In contrast, GNSSo1 and the other monitoring instruments did not record a sustained deformation trend comparable to that observed at GNSSo2. The tiltmeters remained generally stable, except for a minor fluctuation recorded by Tiltmeter o3 in June 2024. The creepmeter data exhibited large and irregular fluctuations, without a clearly identifiable persistent deformation trend. These contrasting responses indicate marked spatial heterogeneity in the monitored deformation and suggest that the active deformation was monitored around the area represented by GNSSo2. Therefore, the rainfall-deformation analysis was primarily based on the GNSSo2 records, which provided the clearest and most continuous deformation signal during the monitored rainfall episodes.

During the June 2024 event, total precipitation reached 716.4 mm, with rainfall highly concentrated between June 9 and June 18. A high-intensity rainfall pulse of 83.3 mm occurred on June 14, followed by an extreme daily rainfall of 193.5 mm on June 15, at which point the early warning was triggered (Fig. 5b). Hourly rainfall data indicate that, following antecedent wetting, a new phase of intense rainfall began at 11:00 on June 15 and peaked at 66.1 mm/h at 12:00 (Fig. 5c). This sustained high-intensity rainfall continued until 18:00 and preceded the reported occurrence of Landslide No. 2 in the early hours of June 16.

Overall, the monitoring data from 2022 and 2024 show that rapid displacement acceleration at GNSSo2 occurred after high-intensity rainfall under antecedently wet conditions. This repeated temporal pattern provided the basis for the subsequent rainfall-deformation analysis.

Landslide No. 2

Landslide No. 2 occurred in the early morning of June 16 following prolonged rainfall. The failure developed within a concave valley and exhibited a larger spatial extent and longer runout than Landslide No. 1. The landslide had an axial length of approximately 300 m and a width of 50–60 m, with an estimated thickness of 3–8 m, a volume of approximately $5.0 \times 10^4 \text{ m}^3$ and a total area of about $1.5 \times 10^4 \text{ m}^2$ (Fig. 6a). The head scarp showed vertical displacement of approximately 5 m (Fig. 6b), and multiple residential structures were damaged along the slope (Figs. 6c and d). The displaced material traveled downslope along a confined channel and accumulated in a low-lying area. Post-failure observations indicated continued creep at the slope toe, with localized ground uplift of 5–10 cm (Fig. 6e).

Field investigations revealed clear morphological differentiation of the initiation, transport, and deposition zones (Figs. 6f–h). The initiation zone was characterized by exposed bedrock and steep slopes. The transport zone exhibited evident lateral erosion and a relatively smooth channelized surface. The deposition zone reached a thickness of approximately 4–5 m and extended over an area of about 6000 m^2 . Cross-sectional profiles (Fig. 6i) further illustrate the internal structure along the landslide path after failure.

At Landslide No. 2, the sliding surface was clearly exposed at the interface between the loose accumulated soil and the underlying weathered bedrock. Field investigations also identified substantial surface runoff traces, blocked drainage ditches, and visible cracks in the upper slope area. These observations suggest possible localized concentration of surface and near-surface water before failure, although no continuous hydrological monitoring was available at Landslide No. 2.

Slope deformation with hydro-meteorological metrics

Figure 7a illustrates the temporal evolution of rainfall parameters (daily and effective rainfall) and cumulative surface displacement recorded by GNSSo1 and GNSSo2 at Landslide No. 1 during the rainfall episodes with significant deformation in 2022 and 2024. The monitoring results reveal a clear temporal association between rainfall and slope deformation, while also indicating an apparent delay between rainfall input and deformation response. Three deformation stages were identified from sustained changes in the slope of the cumulative GNSSo2 displacement curve and the corresponding daily displacement-rate series (Figs. 7a and b). Specifically, the value of daily displacement was derived from the last available hourly record on each day. Stage I was characterized by low displacement rates, generally below approximately 50 mm/day. Stage II was identified by a clear departure from the low-rate condition of stage I and the development of sustained deformation, during which the daily displacement rate increased markedly and typically exceeded approximately 50 mm/day. Stage III was marked by a further steepening of the cumulative displacement curve and rapid acceleration, with displacement rates exceeding approximately 150 mm/day before failure. These values describe the observed rate ranges and were not used as predefined thresholds that had to be satisfied on every day within each stage.

In addition, the relationship between cumulative displacement and effective rainfall shown in Fig. 7b provides further insight into the deformation evolution. During stage I, effective rainfall increased while cumulative displacement remained limited, indicating that rainfall accumulation did not immediately translate into measurable surface deformation. In stage II, cumulative displacement continued to increase even when effective rainfall decreased or stabilized, indicating that the deformation response did not follow rainfall input instantaneously. In stage III, displacement increased sharply following renewed rainfall input, indicating a transition into rapid acceleration.

Figure 7c illustrates the short-term dynamics preceding rapid acceleration based on hourly rainfall and displacement data. A short quasi-stable interval was observed in both monitored events before rapid acceleration, during which rainfall remained elevated while displacement showed little immediate change. This interval was followed by a sharp increase in displacement rate after rainfall intensification. This pattern is consistent with a state-dependent response, in which additional rainfall may have promoted acceleration after the slope had already experienced substantial antecedent wetting and progressive deformation.

Both the 2022 and 2024 events exhibited a short transitional phase between rainfall intensification and rapid deformation. Specifically, 17 June 2022 and 10 June 2024 represented this transition stage, during which effective rainfall increased significantly while

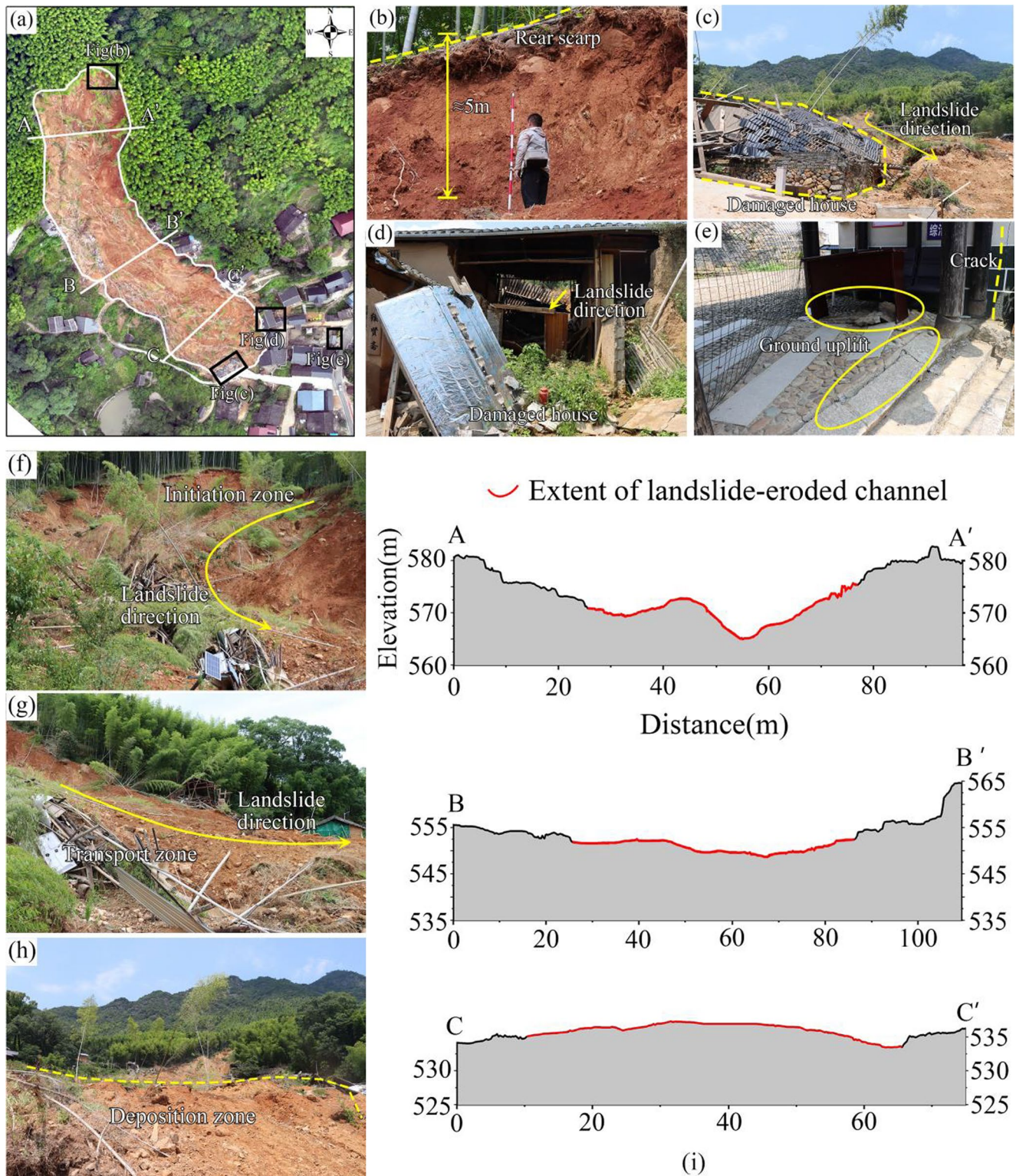


Fig. 6 Characteristics of Landslide No. 2 (photos taken in July 2024): **a** UAV view of the landslide extent; **b** Rear scarp showing approximately 5 m of vertical subsidence; **c–d** damage to residential buildings along the landslide path; **e** Ground uplift of 5–10 cm near the village committee office; **f–h** field views of the initiation, transport, and deposition zones; and **i** transverse profiles along sections A–A', B–B', and C–C'

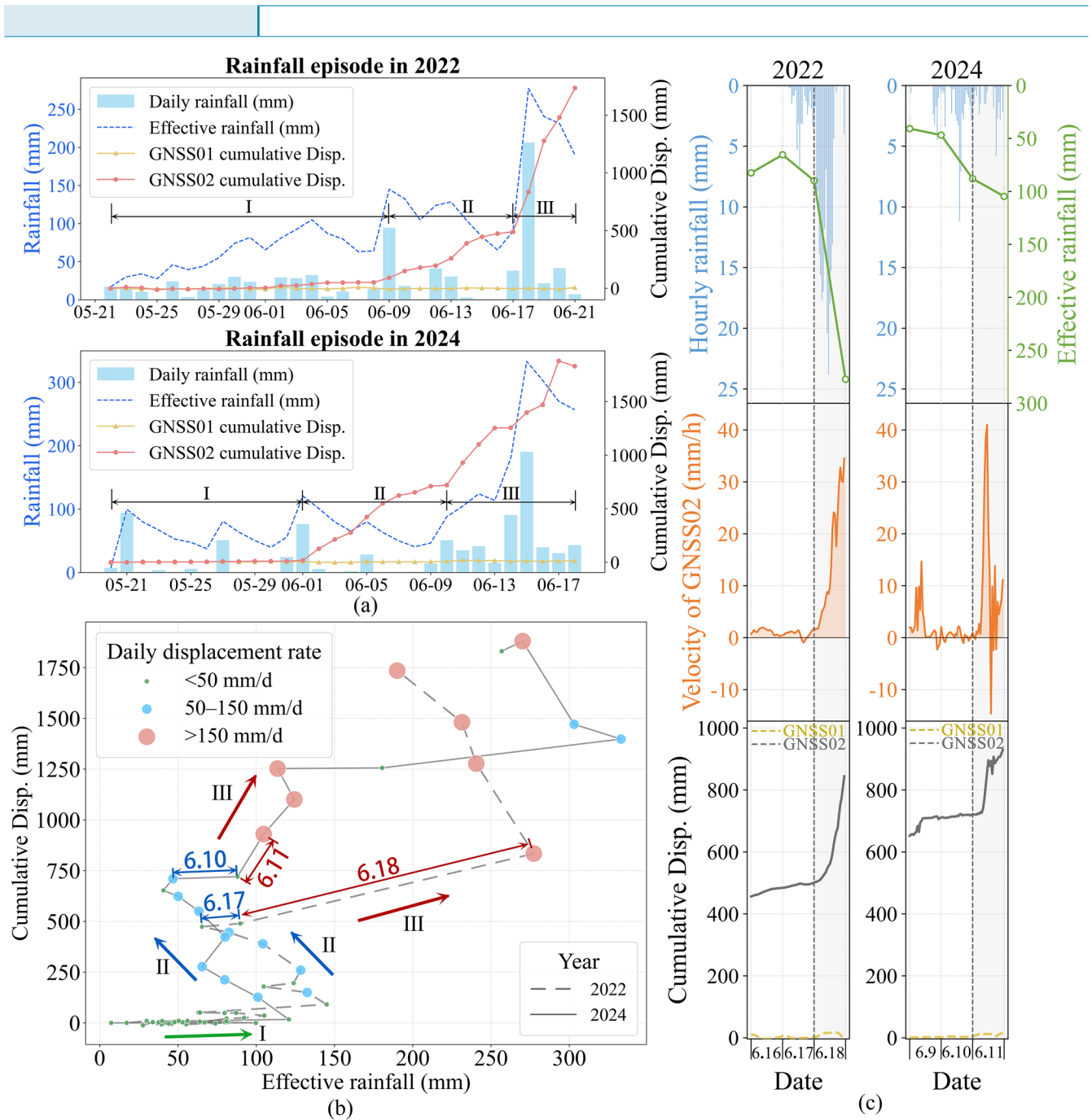


Fig. 7 Analysis of the effective rainfall-cumulative displacement relationship during the rainfall episodes with significant deformation in 2022 and 2024: **a** Time-series monitoring data showing daily rainfall, effective rainfall, and cumulative displacement; **b** Evolutionary path of cumulative displacement versus effective rainfall; **c** Temporal response of hourly displacement characteristics to rainfall intensity during the initial phase of deformation stage III

cumulative displacement remained nearly unchanged (Fig. 7b). The onset of rapid acceleration on 18 June 2022 and 11 June 2024 occurred approximately 1 day after the increase of effective rainfall (Fig. 7c). This observed delayed response suggests that slope deformation was not governed by instantaneous rainfall input alone, but was also related to antecedent rainfall accumulation and the evolving deformation state of the slope. This delay was identified from

only two monitored events and may be affected by data resolution, daily aggregation, and site-specific hydrogeological conditions. It should be regarded as a case-specific observation rather than a fixed hydrological response time or transferable early warning threshold. Because no direct hydro-mechanical observations were available, the underlying processes responsible for the apparent delay cannot be identified directly.

Discussion

Anthropogenic modification of slope hydrological conditions

The displacement evolution described above may have been influenced not only by rainfall forcing but also by site-specific conditions and external modifications. Field surveys identified several forms of anthropogenic disturbance in and around the two landslide sites, including road construction, residential development, deterioration of drainage facilities, and vegetation change (Fig. 8). These

modifications provide important field evidence for understanding the local setting of slope instability and may have increased the sensitivity of the slopes to rainfall by altering slope geometry and local hydrological pathways.

Engineering activities, including road construction and residential development, have produced artificial cut slopes and steep unsupported faces around the landslide sites, thereby potentially increasing local susceptibility to rainfall-induced slope instability (Li et al. 2017; Froude and Petley 2018; Pradhan et al. 2022). These

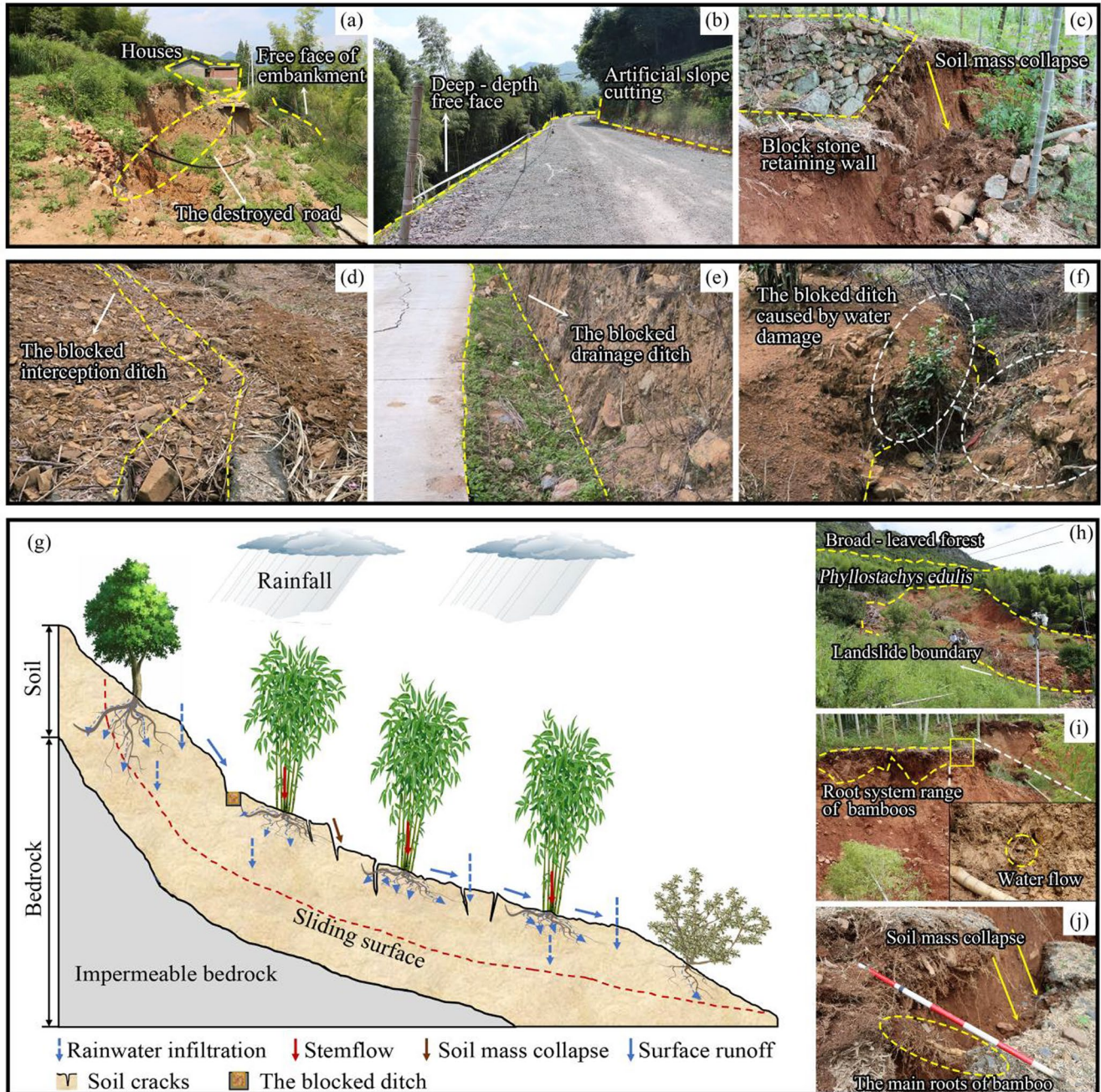


Fig. 8 Influence of human activities on slope hydrology and stability (images taken in July 2024): **a–c** Artificial cut slopes created by engineering activities, increasing slope steepness and instability risk; **d–f** Damaged and clogged drainage facilities due to poor maintenance; and **g–j** Field observations and conceptual schematic of hydrological processes influenced by vegetation

slopes typically exhibited heights of 4–6 m and gradients of 60–75°, locally increasing slope steepness and potentially reducing stability (Fig. 8a). Hence, the formation of local free faces may have further increased local susceptibility (Fig. 8b). At the slope toe, the use of dry-stacked stone retaining walls provided limited engineering reinforcement, particularly under saturated conditions (Fig. 8c). More importantly, the deterioration of drainage infrastructure may have modified local slope hydrology. The original irrigation and drainage system was largely ineffective due to clogging, blockage, and structural damage (Figs. 8d–f). As a result, runoff could not be efficiently discharged and tended to accumulate on the slope. This may promote localized saturation and enhance infiltration during rainfall events (Tu et al. 2019; He et al. 2023).

Vegetation changes further influence slope response through both mechanical and hydrological effects (Ghestem et al. 2011; Bordoni et al. 2016; Vergani et al. 2017; Qin et al. 2022; Murgia et al. 2022; Zhang et al. 2024; DiBiagio et al. 2024; Deng et al. 2025). The replacement of evergreen broad-leaved native forest with *Phyllostachys edulis* plantations had altered root distribution and near-surface water movement pathways. The shallow distribution of *Phyllostachys edulis* roots suggests that their mechanical reinforcement may be limited at the depth of the inferred sliding surface (Tong et al. 2020). Specifically, *Phyllostachys edulis* formed a clear boundary with the adjacent native broad-leaved forest in Landslide No. 2 (Fig. 8h). As depicted in Fig. 8i, the root system of *Phyllostachys edulis* was relatively shallow and was primarily distributed within the upper 1 m of soil, above the inferred sliding surface of Landslide No. 2. Meanwhile, the dense lateral rhizome and fine-root network may modify near-surface soil structure, increase macropore connectivity, and promote lateral subsurface flow or preferential infiltration within the root zone (Lin et al. 2025; Zhao et al. 2025) (Fig. 8g). At Landslide No. 1, field observations indicate that interconnected root systems of *Phyllostachys edulis* form an extensive lateral network (Fig. 8j), which may enhance shallow soil strength and delay crack propagation. At the same time, this connectivity may also facilitate lateral moisture redistribution and contribute to the expansion of the affected area. However, *Phyllostachys edulis*-related effects are discussed here only as possible contributing factors inferred from field observations and previous studies, rather than as directly verified controls on slope instability at this site. Further field experiments and hydrological monitoring would be required to evaluate the specific role of bamboo roots in rainfall infiltration and slope stability.

Overall, these anthropogenic modifications may have increased rainfall sensitivity by altering both local slope geometry and hydrological pathways. Slope cutting, residential development, and limited toe support may have reduced local stability, whereas drainage degradation and vegetation-related root networks may have promoted runoff concentration, preferential infiltration, and subsurface connectivity. These factors should not be interpreted as direct triggers of landslide occurrence because quantitative verification is unavailable. Rather, they may have progressively modified slope conditions, creating a more vulnerable system in which failure may occur under subsequent rainfall events. This highlights the importance of considering both anthropogenic modification and the temporal evolution of slope conditions in rainfall-induced landslide analysis.

Progressive and state-dependent interpretation of landslide occurrence

The combined observations from Landslides No. 1 and No. 2 suggest that landslide occurrence in the study area can be interpreted as a progressive and state-dependent process, rather than as a direct response to individual rainfall events alone. At Landslide No. 1, the 2022 rainfall event produced significant monitored deformation without complete failure, indicating that the slope had entered a sustained deformation state. However, the internal hydromechanical state could not be directly determined from the available monitoring data. The observed deformation may be consistent with progressive internal weakening, including structural disturbance, crack development, and the formation of enhanced pathways for water infiltration (Leroueil 2001; Petley et al. 2005; Yin et al. 2023). This interpretation is consistent with the modified hydrological conditions associated with anthropogenic modifications described above.

In contrast, during the 2024 event, Landslide No. 1 showed localized failure, while Landslide No. 2 experienced rapid, large-scale slope failure. This contrast suggests that slope failure was conditioned not only by rainfall magnitude but also by the antecedent state of the slope. No comparable failure of Landslide No. 2 was reported or observed during the 2022 rainfall event. Although its pre-failure deformation history was not continuously monitored, continued weathering, repeated rainfall, and anthropogenic hydrological modifications may have progressively reduced local slope resistance or increased rainfall sensitivity (Van Asch et al. 1999; Matsuura et al. 2008; Handwerger et al. 2013; Bogaard and Greco 2015). Field evidence, including blocked drainage, concentrated runoff, and crack-assisted infiltration, further suggests that surface water may have been more readily routed into the slope during the 2024 event, potentially promoting infiltration toward the soil–bedrock interface and contributing to failure.

These observations are consistent with a progressive and state-dependent interpretation of landslide occurrence. Under this interpretation, intense rainfall acts as the apparent immediate trigger, while antecedent rainfall accumulation, prior deformation, and locally modified slope conditions influence whether rainfall-induced deformation evolved into failure. Therefore, the transition from rainfall response to failure should be understood not only in terms of rainfall magnitude, but also in terms of the evolving physical and hydrological conditions, and deformation state of the slope.

Implications for landslide monitoring and early warning

The monitoring results indicate that slope deformation was closely associated with cumulative rainfall forcing and delayed displacement response, highlighting the importance of integrating rainfall-derived indicators and displacement response for landslide early warning. From an early warning perspective, these results suggest limitations in relying on rainfall thresholds alone. Effective rainfall was used as an indirect indicator of cumulative rainfall forcing and antecedent hydrological conditions, whereas displacement rate provided a direct measure of the slope response. The combined analysis of effective rainfall and displacement data can help identify

the transition from limited deformation to sustained deformation and then rapid acceleration, although the underlying hydrological processes require further verification using direct subsurface monitoring. The transition from sustained deformation (stage II) to rapid acceleration (stage III) therefore represented a potentially useful alerting phase. The quasi-stable interval preceding acceleration, as observed in Fig. 7c, may represent an alerting interval rather than a deterministic early-warning window. In this case, the relevant transition dates were identified as 17 June for the 2022 event and 10 June for the 2024 event, providing a limited lead time in these two monitored events. In a real-time monitoring context, this transitional rainfall-deformation pattern can be identified when effective rainfall remains elevated relative to the preceding background level while the displacement rate shows a sustained deviation from its site-specific background behavior, even before rapid displacement acceleration becomes evident.

It should also be noted that similar rainfall-deformation patterns may occur without immediate failure. A temporary increase in effective rainfall and displacement rate may represent transient hydrological loading or non-progressive deformation, particularly when the slope has not yet developed into a progressively accelerating state. Therefore, this pattern should not be used as a stand-alone failure warning criterion. Instead, it may serve as an alerting condition for intensified monitoring, field inspection, and reassessment of the local hazard situation. For operational early warning, a comprehensive assessment should be conducted based on the site-specific rainfall threshold, short-term rainfall forecast, as well as field evidence such as crack expansion, toe deformation or drainage blockage. Additional monitored events, including non-failure rainfall episodes, are required to evaluate the false-alarm potential and to test whether the transition pattern observed in this study can be generalized to other landslides or rainfall episodes.

Conclusions

This study investigated two rainfall-induced landslides in Wushantou Village using remote sensing, field surveys, in situ monitoring, geotechnical investigations, and laboratory testing. The two landslides showed distinct deformation and failure characteristics. Landslide No. 1 exhibited significant monitored deformation in 2022 and localized failure in 2024, whereas Landslide No. 2 experienced rapid, large-scale failure in 2024 along the soil-bedrock interface, with field observations indicating concentrated runoff and crack-assisted infiltration.

The observed deformation at Landslide No. 1 showed a stage-based evolution, including an initial stage with limited response, a sustained deformation phase, and a final acceleration stage. The onset of rapid acceleration occurred approximately 1 day after the increase of effective rainfall. Overall, the slope deformation was closely associated with cumulative rainfall forcing and delayed displacement response, highlighting the importance of integrating rainfall-derived indicators and displacement response for landslide early warning.

Anthropogenic modifications, including slope cutting, drainage degradation, and vegetation change, may have increased rainfall sensitivity by altering both local slope geometry and hydrological pathways. These findings suggest that landslide failure in the study area was associated not only with rainfall magnitude, but also with antecedent rainfall accumulation, deformation state, and local slope conditions. Future work should incorporate direct monitoring of

groundwater level, pore-water pressure, soil moisture, or matric suction to better constrain the underlying hydrological mechanisms and evaluate the applicability of this stage-based interpretation across different geological and climatic settings.

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Data availability

The data are available from the corresponding authors upon reasonable request.

Declarations

Competing interests The authors declare no competing interests.

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